CENTRE FOR DEVELOPMENT OF IMAGING TECHNOLOGY (C-DIT)

# BHASHAMITHRAM II

## ഭാഷാമിത്രം - II നിഘണ്ടു മൊബൈൽ ആപ്പ്

Comprehensive Technical Specification, Feature Catalogue & Architecture Report

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|---|---|---|---|
| **Application ID** | org.cdit.bhashamithram2 | **Production Version** | v1.0.7 (VersionCode 9) |
| **Project Name** | Bhashamithram II Mobile App | **Target Framework** | React Native 0.86.3 / Expo SDK 57 |
| **Operating System** | Android 15 (API 36) Compliant (Min SDK 24) | **Database Architecture** | Local Offline SQLite (~823,000 Entries) |
| **Linguistic Streams** | 5 Dictionaries (Eng-Mal, Mal-Eng, Synonyms, etc.) | **Release Package** | AAB (293 MB Offline Lexicon) |
| **Institution** | C-DIT, Government of Kerala | **Documentation Date** | October 2026 |

**Executive Summary:** Bhashamithram II is Kerala's flagship bilingual lexicographical mobile application engineered by the Centre for Development of Imaging Technology (C-DIT). Operating completely offline via an embedded, high-performance SQLite database of over 823,000 lexical records, the application bridges the linguistic divide with five multi-directional dictionaries, voice search with intelligent intent parsing, real-time Manglish-to-Malayalam transliteration, speech synthesis, and crowdsourced lexical additions.

# 1. High-Level Architecture & Technical Paradigm

Bhashamithram II is engineered as an **Offline-First, Cross-Platform Mobile Application** built upon the modern React Native (v0.86.3) and Expo SDK 57 architecture. The core design ensures that all essential dictionary query capabilities function completely without an internet connection, allowing students, researchers, journalists, and language enthusiasts in low-connectivity areas to access exhaustive Malayalam lexical resources instantly.

| Architectural Layer | Technology / Component | Technical Role & Description |
|---|---|---|
| **Presentation & UI** | React Native 19, Expo Router ~57.0.21, React Native Reanimated 4.5.1 | File-based routing, declarative UI components, fluid gesture handlers, dynamic floating Malayalam letters canvas background, and responsive layout engine across phones and tablets. |
| **Data & Offline Storage** | expo-sqlite ~57.0.3, @react-native-async-storage/async-storage | Embedded 290MB SQLite database (dictionary_v2.db) with 823,000+ records, automated asset import, self-healing corruption recovery, and persistent user bookmarks and recent activity. |
| **Multimodal & Audio** | expo-speech ~57.0.3, expo-audio ~57.0.5, Native SpeechRecognizer | Bilingual speech-to-text with conversational intent extraction, silence auto-detection, and offline text-to-speech audio pronunciation with native speed/pitch control. |
| **Transliteration Engine** | Custom Pure-TS Manglish Transliterator | Real-time phonological Latin-to-Malayalam transliteration mapping engine running at 0ms latency with zero network overhead. |
| **Security & Sanitation** | Custom Cryptographic Security Module (security.ts, cacheManager.ts) | FIPS PUB 180-4 SHA-256 and HMAC-SHA256 signatures, strict XSS and SQL injection filters, and OWASP MASVS MSTG-STORAGE-2 cache auto-purging. |
| **Backend & Cloud Sync** | Node.js, Express 5.2.1, Sequelize 6.37.8, MySQL2 | RESTful microservice for crowdsourced lexical submissions, committee approval workflows, administrative review portal, and cloud database synchronization. |
| **Native Build Toolchain** | AGP 8.12.0, Gradle 9.3.1, Kotlin 2.1.20, R8 Full Mode | Android 15 (API 36) edge-to-edge system bar compliance, dead-code stripping, optimized resource shrinking, and native C++ autolinking. |

# 2. Comprehensive Feature Catalogue & Functional Modules

## 1. English - Malayalam Dictionary (ഇംഗ്ലീഷ് - മലയാളം നിഘണ്ടു)

Enables exhaustive translation of English headwords into Malayalam. Provides multiple nuanced meanings, contextual sentence usage, phonetic pronunciation guides, grammatical parts of speech, and comparative Dravidian equivalents (Tamil, Kannada, Telugu, and Tulu). Features real-time prefix search and indexed sub-millisecond retrieval.

## 2. Malayalam - English Dictionary (മലയാളം - ഇംഗ്ലീഷ് നിഘണ്ടു)

Allows looking up Malayalam words (typed via Malayalam keyboard or generated via Manglish transliteration) to find authoritative English translations, etymological roots, grammatical gender, inflections, and contextual definitions.

## 3. Malayalam Synonyms & Polysemy (മലയാളം നാനാർത്ഥ നിഘണ്ടു)

A specialized lexicographical module detailing polysemous words (Nanartham). Unpacks words that hold multiple divergent meanings across classical literature, poetry, legal terminology, and modern colloquial usage.

### 4. Malayalam - Malayalam Monolingual Dictionary (മലയാളം - മലയാളം നിഘണ്ടു)

An authoritative monolingual dictionary built on canonical references (such as Gundert's and Sabdatharavali traditions). Includes traditional proverbs (Pazhamchol), regional dialectal variations across Kerala, and cultural context notes.

### 5. English - English Dictionary

Monolingual English lexical definitions, phonetics, inflected forms, and grammatical classifications designed for students and writers needing precise bilingual cross-referencing within the same application.

### 6. Crowdsourced Lexicon & Suggest-a-Word (വാക്ക് നിർദ്ദശിക്കുക)

A public contribution interface allowing language teachers, researchers, and users to propose missing words, modern scientific neologisms, and regional idioms. Includes an interactive multi-tab tracking screen: എന്റെ നിർദ്ദശങ്ങൾ (My Submissions), സമിതി നിർദ്ദശങ്ങൾ (Committee Review), സ്വീകരിച്ച നിർദ്ദശങ്ങൾ (Approved), and നിരസിച്ച നിർദ്ദശങ്ങൾ (Rejected). Synchronizes with local SQLite and remote REST backend.

### 7. Intelligent Bilingual Voice Search (ശബ്ദ തിരച്ചിൽ)

State-of-the-art voice recognition engine utilizing the native Android SpeechRecognizer with automated speech break detection and auto-stop. Incorporates an algorithmic query cleaner that strips conversational phrases ('search for...', 'meaning of...', 'വാക്കിന്റെ അർത്ഥം...') to isolate the exact target headword in either language.

### 8. Text-to-Speech & Pronunciation (ശബ്ദ ഉച്ചാരണം)

Integrated speech synthesis engine via expo-speech and expo-audio. Synthesizes clear audio pronunciations for English and Malayalam words with customizable speech rate, volume, and acoustic pitch.

### 9. Offline Manglish-to-Malayalam Transliteration Engine

High-speed phonological transliterator allowing users without a native Malayalam keyboard to type in English script (e.g., 'namaskaram', 'pusthakam', 'bhashamithram') and have it automatically converted into Malayalam Unicode script ('നമസ്കാരം', 'പുസ്തകം', 'ഭാഷാമിത്രം') in real time without network connectivity.

### 10. Favorites & Recent Search History Management

Persistent tracking of starred/bookmarked words and recent query history. Features instant search reactivation, single-word deletion without intrusive confirmation alerts, and a bulk-clear option with state consistency.

### 11. Institutional Portal & Direct Play Store Integration

Dedicated information and support screen detailing C-DIT's technological mission, government credentials, developer credits, direct deep-linking to the official Google Play Console listing (org.cdit.bhashamithram2) for ratings and updates, and native share dialogs.

# 3. Complete Package, Library & Framework Inventory

Every dependency in Bhashamithram II has been audited for security, performance, minimal runtime overhead, and strict Android 15 compatibility. Below is the exhaustive inventory of all libraries powering the application.

## A. Core Runtime & Expo SDK Ecosystem

| Package Name | Version | Category | Technical Purpose & Role |
|---|---|---|---|
| **react** | 19.2.3 | Core Framework | Underlying component rendering, concurrent mode, hooks architecture, and memory lifecycle. |
| **react-native** | 0.86.3 | Mobile Platform | React Native engine with New Architecture JNI autolinking, Fabric renderer, and bridge communication. |
| **expo** | ~57.0.23 | Application Framework | Expo SDK 57 unified toolchain, native runtime configurations, and module lifecycle management. |
| **expo-router** | ~57.0.21 | Navigation | Universal filesystem-based routing architecture with typed routes, deep linking, and stack transitions. |
| **expo-sqlite** | ~57.0.3 | Local Storage | High-performance native SQLite engine with WAL mode, prepared statement batching, and asset importing. |
| **expo-speech** | ~57.0.3 | Audio & Voice | Native Text-to-Speech synthesis binding to Android TTS engines for English and Malayalam locale pronunciations. |
| **expo-audio** | ~57.0.5 | Audio System | Audio recording, buffer management for speech fallback, and patched background notification service. |
| **expo-image** | ~57.0.5 | Media Loading | Glide-backed high-performance image rendering engine with memory caching and downsampling. |
| **expo-font** | ~57.0.1 | Typography | Custom font loader handling Noto Sans Malayalam variable font asynchronously before layout mount. |
| **expo-splash-screen** | ~57.0.9 | Startup & Splash | Controls native splash lifecycle, coordinate pre-draw listeners, and manages zero-flicker transitions. |
| **expo-status-bar** | ~57.0.1 | System Bar | Configures edge-to-edge transparent status bars across Android 15 and iOS devices. |
| **expo-file-system** | ~57.0.1 | File Management | Accesses app-private storage, sandboxed cache directories, and database asset streams. |
| **expo-constants** | ~57.0.18 | App Metadata | Provides runtime manifest inspection, package name, build versions, and device characteristics. |
| **expo-linking** | ~57.0.10 | Deep Linking | Handles intent URL schemes ('exp+bhashamithram2://', 'https://play.google.com/...'). |
| **expo-build-properties** | ~57.0.20 | Build Configuration | Injects custom Gradle build configurations, minSdk 24, compileSdk 36, and ProGuard rules. |
| **expo-system-ui** | ~57.0.4 | System UI | Controls root Android view window backgrounds and dark/light system theme adaptations. |

## B. UI, Gesture, Animation, Native & Development Tools

| Package Name | Version | Category | Technical Purpose & Role |
|---|---|---|---|
| **react-native-reanimated** | 4.5.1 | Animation | Runs 60FPS fluid animations on the native UI thread; drives the floating Malayalam letters background canvas. |
| **react-native-gesture-handler** | ~2.32.0 | Touch Interaction | Native touch recognizers for off-canvas drawer navigation, card swipes, and search interactions. |
| **react-native-screens** | ~4.26.0 | Navigation Views | Native Android Fragment & Activity view-hierarchy containers for optimal memory management. |
| **react-native-safe-area-context** | ~5.7.0 | Viewport Insets | Calculates dynamic device insets for display cutouts, notches, navigation gestures, and Android 15 edge-to-edge. |
| **@react-native-async-storage** | 2.2.0 | Key-Value Cache | Persistent offline key-value storage for app settings, database version tokens, and user preferences. |
| **@expo/vector-icons** | ^15.0.2 | Iconography | Vector icon suite (Ionicons, MaterialCommunityIcons, Feather) optimized into native glyph fonts. |
| **patch-package** | ^8.0.1 | Engineering Tool | Persists native Java/Kotlin library patches (e.g. expo-audio network decode patch) across clean installs. |
| **typescript** | ~6.0.3 | Language | Enforces static typing, strict null safety, and clean interfaces across services and components. |
| **Kotlin Compiler** | 2.1.20 | Native Language | Compiles native Android Kotlin modules and React Native Gradle plugin components. |
| **Android Gradle Plugin (AGP)** | 8.12.0 | Build System | Android compilation, R8 code minifier, optimized resource shrinker, and APK/AAB packager. |
| **Gradle Wrapper** | 9.3.1 | Build Automation | High-performance multi-threaded native build daemon with daemon memory metaspace configuration. |
| **Android NDK** | 27.1.12297006 | C/C++ Toolchain | Compiles CMake native JNI shared libraries (.so) for Reanimated, Worklets, and React Android. |

## C. Backend Microservices & Cloud Infrastructure

| Package / Service | Version | Category | Technical Purpose & Role |
|---|---|---|---|
| **express** | ^5.2.1 | API Framework | RESTful microservice handling crowdsourced word proposals, validation, and committee moderation. |
| **sequelize** | ^6.37.8 | ORM Layer | Object-Relational Mapping for relational schema management, models, transactions, and migrations. |
| **mysql2** | ^3.22.6 | Database Driver | High-performance asynchronous MySQL client driver supporting connection pooling and secure SSL channels. |
| **@google-cloud/speech** | ^7.5.0 | Cloud Speech AI | Google Cloud Speech-to-Text API client for high-accuracy Malayalam audio transcription and intent parsing. |
| **cors & dotenv** | ^2.8.6 / ^17.4.2 | Security & Config | Cross-Origin Resource Sharing policy enforcement and secure environment variable management. |
| **sqlite3** | ^6.0.1 | Local DB Engine | C/C++ SQLite3 native bindings for database asset verification, staging, and pre-compilation. |
| **Render Cloud PaaS** | Production | Hosting Infrastructure | Containerized cloud hosting with automated HTTPS, health monitoring, and zero-downtime deployment pipelines. |

# 4. Database Engineering & Offline Lexical Dataset

At the core of Bhashamithram II is a pre-indexed, relational SQLite database (**dictionary_v2.db**) bundled directly into the application assets. This architecture enables instant, high-concurrency offline lookups without server latency or connectivity requirements.

| Database Metric | Metric Value | Engineering Rationale |
|---|---|---|
| **Total Lexical Entries** | 823,540 total rows | Consolidated across headwords, multi-category meanings, and etymological roots. |
| **Table: 'dictionary'** | 233,464 headwords | Primary dictionary headwords indexed with phonetic transcription, grammatical category, and cognates. |
| **Table: 'meanings'** | 551,093 definitions | Normalized relational table linking contextual usages and multiple shades of meaning per dictionary ID. |
| **Table: 'dictionary_roots_backup'** | 38,983 roots | Linguistic root terms preserving classical morphological and Sanskrit/Dravidian etymological derivations. |
| **Database Asset Size** | ~290 MB embedded | Embedded in the release bundle; extracted into app-private SQLite storage upon first launch. |
| **Primary Database Indices** | idx_word_type, idx_dict_type_word | Compound B-Tree indices ensuring exact and prefix queries execute in under 1 millisecond. |

## A. Dictionary Headword Schema Attributes

Each record in the primary dictionary table supports 35 structured lexicographical fields:

- **Core Identification:** id (Auto Primary Key), word (Indexed Headword string), dictionary_type (1=Eng-Mal, 2=Mal-Eng, 3=Synonyms, 4=Mal-Mal, 5=Eng-Eng).
- **Phonology & Morphology:** phonetic_transcription (IPA representation), root (Morphological root), etymology (Historical derivation), inflections (Grammatical conjugations).
- **Comparative Dravidian Philology:** tamil_equivalent, kannada_equivalent, telugu_equivalent, tulu_equivalent, english_equivalent.
- **Cultural & Literary Markers:** cultural_note (Heritage significance), dialects (Regional Malayalam dialect markers: Malabar, Travancore, etc.), proverb (Traditional Pazhamchol sayings).
- **Lexical Relations:** synonyms (Comma-separated synonyms), antonyms (Opposites), similar_word, novel_words (Modern coined terms).
- **Governance & Review:** approved, superuser_approved, approved_at, submitted_by_id, from_suggestion_id.

## B. Self-Healing Native Database Asset Pipeline

The native SQLite initialization layer (database.native.ts) incorporates an automated self-healing asset lifecycle. When the application launches, the system evaluates the bundled database asset version token (@dictionary_db_version). If the local SQLite database file is missing, empty, or outdated following an app upgrade, importDatabaseFromAssetAsync automatically overwrites and re-indexes the database in private storage. A verification query executes against sqlite_master before opening the connection to guarantee zero runtime database crashes.

# 5. User Interface, Theming & Responsive Design System

• **Official Bhashamithram-II Design Language:** Features a serene light pastel palette (#F8FAFC) accented with deep ocean blue typography (#0B3B60), mint/teal headers (#0D9488), and botanical leaf elements (#16A34A). All screens adhere to a calm, readable aesthetic inspired by Kerala's lush natural landscape.

• **Interactive Floating Malayalam Letters Canvas:** The custom component FloatingLettersBackground.tsx renders an animated ambient background displaying Malayalam vowels and consonants (അ, ആ, ഇ, ഈ, ഉ, എ, ഒ, ക, ഗ, ങ, ച, ണ, ത, പ, മ, യ, ര, ല, വ, ശ, ഷ, സ, ഹ) floating with gentle harmonic oscillations behind content cards to celebrate linguistic identity.

• **Multi-Tier Responsive Viewport Engine:** Utilizes the centralized useResponsive() hook to adapt layout dimensions dynamically across four device classes: Small Phones (<360dp width), Standard Phones (360–599dp), Tablets (600–1023dp), and Large Screens (≥1024dp). Emblem dimensions, typography scale, padding insets, and card heights adjust fluidly to prevent clipping.

• **Zero-Flicker Native Splash & Loading Lifecycle:** Engineered with a coordinated startup lifecycle: the native Android splash screen displays the official Bhashamithram II branding lockup, which dismisses synchronously the moment React Native mounts, revealing the clean branded loading page without spinners, before presenting the main dashboard.

# 6. Security, Privacy & OWASP MASVS Compliance

Bhashamithram II follows strict enterprise-grade security standards to protect user privacy and prevent client-side tampering or injection vulnerabilities:

• **OWASP MASVS MSTG-STORAGE-2 (CWE-276) Cache Management:** The utility cacheManager.ts automatically clears memory and disk image caches and scrubs the app-private cache directory (context.cacheDir) on session initialization. Temporary audio recordings and scratch files are purged immediately to eliminate data remanence.

• **Pure TypeScript Cryptographic Signatures (security.ts):** Implements high-performance FIPS PUB 180-4 SHA-256 and RFC 2104 HMAC-SHA256 hashing in pure TypeScript. Enables cryptographic payload signing for crowdsourced word submissions without relying on third-party binary modules.

• **Multi-Stage Input Sanitization:** Protects against Cross-Site Scripting (XSS), SQL injection, and buffer manipulation. Form inputs undergo real-time alphabet filtering, stripping dangerous delimiters, HTML tags, and unescaped quotes while supporting Unicode Malayalam.

• **Hardened Android Application Manifest:** Configured with android:allowBackup='false' to prevent unauthorized backup extraction, android:usesCleartextTraffic='false' to enforce HTTPS encryption, and explicitly blocked dangerous permissions such as SYSTEM_ALERT_WINDOW.

# 7. Android 15 & Google Play Console Engineering Optimizations

During pre-launch inspections for release build 9 (v1.0.7), the codebase was updated to resolve all Google Play Console recommendations and ensure future-proof stability on Android 15 (Target SDK 36):

| Optimization Area | Play Console Finding | Resolution & Engineering Fix |
|---|---|---|
| **Android 15 Edge-to-Edge Enforcement** | Deprecated window color APIs (setStatusBarColor, setNavigationBarColor). | Removed deprecated XML tags from styles.xml; added values-v35/styles.xml with windowOptOutEdgeToEdgeEnforcement=false; added ProGuard rules to strip runtime status bar tint invocations. |
| **Bitmap Stream Memory Optimization** | Un-downsampled network stream bitmap decodes causing OOM risks. | Patched expo-audio's AudioControlsService to stub unneeded remote notification artwork loading (saved permanently via patch-package); disabled unused Fresco GIF and WebP native decoders in gradle.properties. |
| **R8 Full Mode & Resource Shrinking** | Optimized resource shrinking and R8 full mode not enabled. | Enabled android.enableR8.fullMode=true and android.r8.optimizedResourceShrinking=true; cleaned proguard-rules.pro from overly broad keep rules, allowing R8 to dead-code strip 15+ MB of unused classes. |
| **React Native Codegen Clean Resilience** | CMake clean failure: 'add_subdirectory given source codegen/jni which is not an existing directory'. | Configured android/app/build.gradle to disable externalNativeBuildClean tasks and clean .cxx directly, enabling flawless execution of './gradlew clean bundleRelease'. |

# 8. Production Build & Release Artifact Specifications

| Artifact Specification | Production Value / Configuration |
|---|---|
| **Package Name / Application ID** | org.cdit.bhashamithram2 |
| **Version Name & Version Code** | Version 1.0.7 (VersionCode 9) |
| **Production Release Bundle Path** | android/app/build/outputs/bundle/release/app-release.aab |
| **Android App Bundle (AAB) Size** | 293 MB (Encompasses 823,000+ offline lexical records) |
| **Minimum Supported Android Version** | Android 7.0 Nougat (API Level 24) |
| **Target & Compile Android SDK** | Android 15 (API Level 36) with Build Tools 36.0.0 |
| **Target Architectures** | arm64-v8a, armeabi-v7a, x86, x86_64 |
| **R8 Code Minifier & Shrinker** | Enabled (Full Mode + Optimized Resource Shrinking) |
| **Google Play Store Status** | Ready for Play Console production rollout |

**Institutional Accreditation:**
This technical documentation reflects the complete architectural specification and feature catalogue of the Bhashamithram II Mobile Application project, developed under the Research & Development initiative of the **Centre for Development of Imaging Technology (C-DIT)**, Thiruvananthapuram, Government of Kerala.
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